

next stage formed by transistor T2 and inductor L2, and finally to the air through a piano-wire antenna.

The output is AC coupled through capacitor C5 to remove any remaining DC components. The L3-C6 network

is for impedance matching of the antenna. The 27MHz transmitter is turned on/off by the data (and control) signal from IC1. Because of this active state the output is sinewave and the spacing between on/off transitions is in tune with the signal input from the encoder chip.

Circuit diagram of the receiver in 27MHz toy car is shown in Fig. 3. Circuit of the 4.5V battery (three 1.5V AA cells) powered radio receiver board QF-1688-R3 located inside the toy car is a bit more complicated as it also includes integrated H-bridges for driving the two DC motors. Core of the radio receiver board is 16-pin decoder chip TXM8D423C (IC1).

Front end of the radio receiver circuit is in fact a regenerative receiver that has been in use for decades. It is built around transistor T1. A tuned LC circuit (tank circuit) comprising L1 and C2 provides the positive (regenerative) feedback but only at the tuned frequency of 27MHz.

Output signal from the tuned LC circuit is filtered to remove the carrier so that only data signal reaches IC1. Two standard H-bridge circuits inside IC1 are used for driving the propulsion motor for back and front (B/F) movement and the steering motor for left and right (R/L) movement. The H-bridges allow smooth bidirectional control of the DC motors.

Fig. 4 shows the transmitter and receiver modules that are now available from many online shops as toy-grade replacement parts.

You can assemble the transmitter and receiver circuits on a general-purpose PCB. However, since 27MHz transmitter and receiver modules are now easily available in the market, their use would be a better option.

Connect the 3V battery to the transmitter module and enclose it in a toy car remote. Then connect the motors to the receiver module as shown in Fig. 4. Enclose the receiver in a toy car body. **EFY**

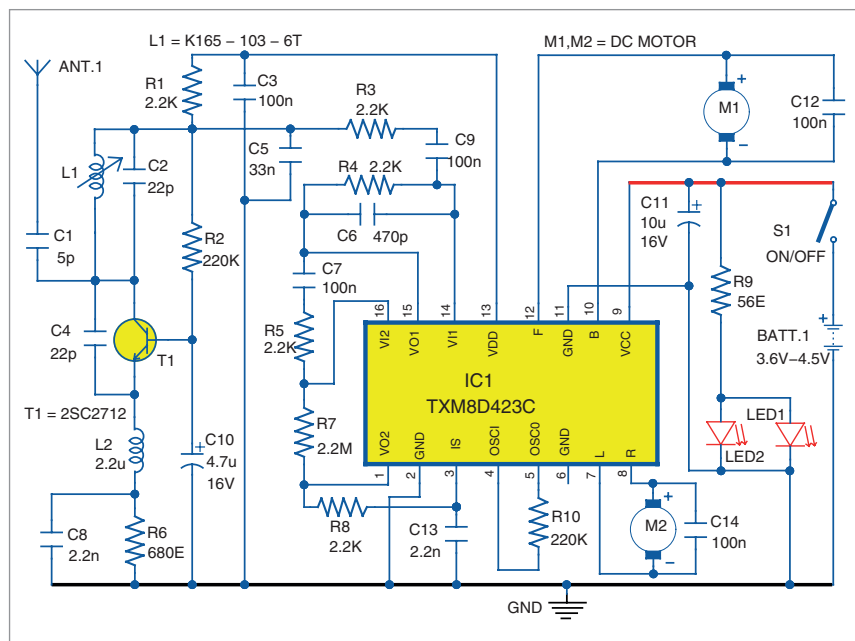


Fig. 3: Circuit diagram of the receiver

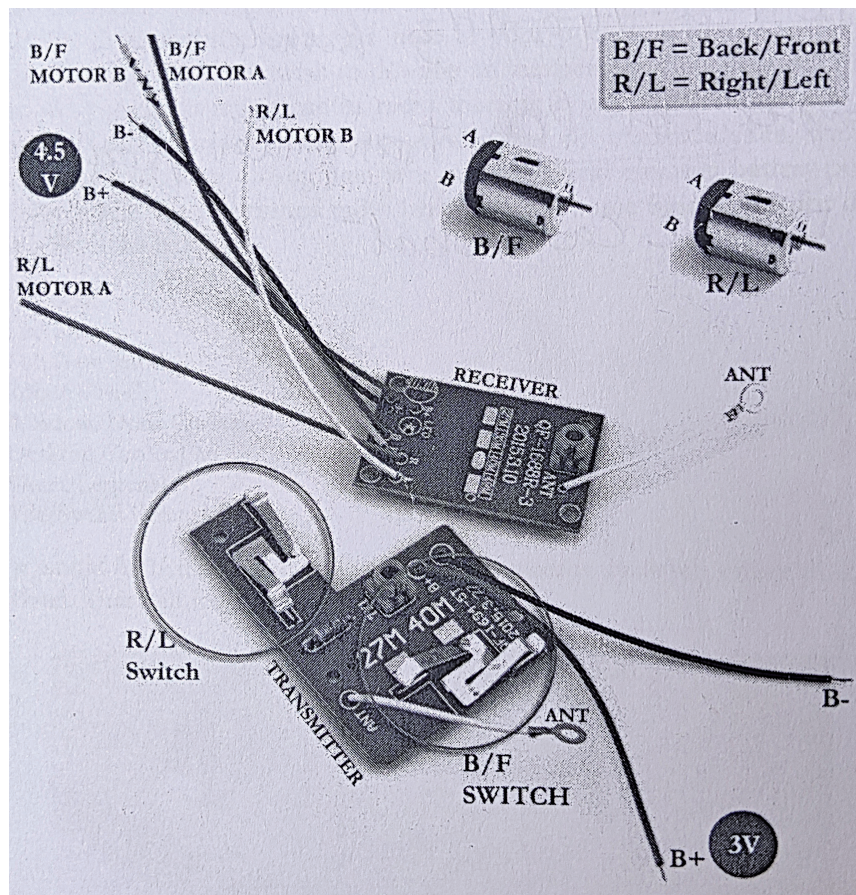


Fig. 4: Transmitter and receiver modules with pin details